

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO4: Evaluate recognition and learning methods including machine learning and deep learning models (CNNs, GANs, SVM, HMM, etc.) for vision tasks.. (BL5)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A

Data Interpretation

Code of Conduct:

*I P Sai Bhavyatha(192424333)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Palacharla Sai Bhavyatha

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Annexure A

Data Interpretation

Question 1: Hough Transform & Shape Detection

Scenario: Detect linear features (roads, rivers) in satellite images.

Input Data:

- A. High-resolution grayscale satellite images of urban and rural areas (e.g., 1024×1024 pixels, TIFF or PNG)
- B. Images contain noise (clouds, shadows) and partial occlusions
- C. Multiple linear features (roads, rivers, canals) with intersections
- D. Optional synthetic dataset: images with drawn lines at random angles and lengths to test Hough Transform

Question 2: PCA & Shape Priors for Object Recognition Scenario:

Segment organs in MRI scans using shape priors. **Input Data:**

- A. MRI slices (grayscale, 512×512 pixels) of organs like liver or heart
- B. Training set: 50–100 annotated organ masks for PCA-based shape model construction
- C. Test set: 10–20 MRI images with partial organ boundaries or noise
- D. File formats: DICOM or PNG

Question 3: Machine Learning Methods (SVM, HMM, GMM, EM)

Scenario: Human activity recognition from video feature trajectories. **Input**

Data:

- A. Video sequences (30–60 FPS) of activities like walking, running, jumping (e.g., 640×480 pixels, MP4)
- B. Feature trajectories extracted per frame: joint positions (x, y coordinates) or optical flow vectors
- C. Training labels: activity per video (e.g., walking, jumping)
- D. Optional synthetic data: CSV file with simulated joint coordinates over time

Question 4: Deep Learning (CNNs, Transformers, GANs) Scenario:

Detect rare obstacles in urban self-driving car images. **Input Data:**

- A. RGB camera images of urban streets (1024×768 pixels, JPG/PNG)
- B. Labeled dataset with bounding boxes for obstacles (rare objects, e.g., traffic cones, animals)

- C. Small dataset: 500–1000 labeled images for real obstacles
- D. Optional synthetic images generated by GAN for data augmentation

Question 5: Graph-Based Learning for Shape Correspondence

Scenario: Find correspondences between 3D human face meshes. **Input**

Data:

- A. 3D meshes of human faces (OBJ or PLY format) with varying vertex counts (5,000–50,000 vertices)
- B. Training set: 20–50 face meshes with annotated landmark correspondences
- C. Test set: 5–10 unseen face meshes
- D. Optional: Graph adjacency matrices representing each mesh for input to graph-based algorithms

ANSWERS

Question 1: Hough Transform & Shape Detection

Scenario

Detect linear features such as roads, rivers, and canals in satellite images using the Hough Transform.

Answer

Understanding of Data:

The input consists of high-resolution grayscale satellite images of urban and rural areas with a typical size of 1024×1024 pixels. The images may contain noise caused by clouds, shadows, vegetation, and other objects. Roads, rivers, and canals appear as linear structures, but some portions may be partially occluded. Intersections between different linear features can also occur. The objective is to identify these linear patterns accurately.

Analysis:

The Hough Transform is suitable for detecting straight lines even when the lines are incomplete or affected by noise. First, the grayscale image is converted into an edge image, commonly using an edge detection technique such as Canny Edge Detection. The Hough Transform then represents each possible line using the parameters ρ (distance from the origin) and θ (angle of the line). Edge points vote for possible lines in the Hough parameter space. Strong peaks in the accumulator indicate the presence of significant linear features.

For example, if several edge pixels belong to the same road, their votes accumulate

around the corresponding ρ and θ values. This allows the road to be detected even when some pixels are missing because of shadows or partial occlusion.

Application of Concepts:

The following procedure can be applied:

1. Read the satellite image.
2. Convert the image into grayscale if required.
3. Apply noise reduction using Gaussian filtering.
4. Perform Canny edge detection.
5. Apply the Hough Line Transform.
6. Identify peaks in the Hough accumulator.
7. Convert the detected parameters into line segments.
8. Draw the detected roads, rivers, and canals on the original image.
9. Compare the detected lines with the actual linear features.

The Hough Transform can be evaluated using the number of correctly detected lines, false detections, and missed lines.

Conclusion:

The Hough Transform provides an effective method for detecting linear structures in satellite images. It is particularly useful when roads, rivers, or canals are partially interrupted by noise or occlusion. Proper preprocessing improves detection accuracy. However, curved roads and rivers may require additional techniques because the standard Hough Transform is mainly designed for straight-line detection.

Question 2: PCA & Shape Priors for Object Recognition

Scenario

Segment organs such as the liver or heart in MRI scans using PCA-based shape priors.

Answer

Understanding of Data:

The input consists of grayscale MRI slices of approximately 512×512 pixels. A training set of 50–100 annotated organ masks is available for constructing a statistical shape model. The test set contains approximately 10–20 MRI images where organ boundaries may be incomplete, noisy, or unclear. The objective is to segment the organ accurately by using both image information and prior knowledge about its shape.

Analysis:

Principal Component Analysis (PCA) can be used to identify the major variations in organ shapes within the training dataset. First, all annotated organ masks are aligned to a common coordinate system. Each shape is represented as a vector of landmark coordinates or boundary points. PCA is then applied to these vectors to obtain the mean organ shape and the principal modes of variation.

The mathematical representation of a shape can be expressed as:

$$\mathbf{S} = \bar{\mathbf{S}} + \mathbf{P}\mathbf{b}$$

where:

- $\mathbf{S}$ = reconstructed organ shape
- $\bar{\mathbf{S}}$ = mean organ shape
- $\mathbf{P}$ = principal component matrix
- $\mathbf{b}$ = shape parameters

The principal components describe the most important ways in which the organ shape varies among different patients.

Application of Concepts:

The following procedure can be followed:

1. Collect annotated MRI organ masks.
2. Extract the organ boundaries.
3. Align the training shapes.
4. Convert each shape into a numerical vector.
5. Calculate the mean shape.
6. Apply PCA to the training shapes.
7. Retain the principal components explaining most of the variance.
8. Construct the statistical shape model.
9. Apply the model to test MRI images.
10. Combine image boundaries with the PCA shape prior.

11. Obtain the final organ segmentation.
12. Compare the predicted mask with the ground-truth mask.

Performance can be measured using **Dice Similarity Coefficient, Intersection over Union (IoU), precision, and recall.**

Conclusion:

PCA-based shape priors improve organ segmentation when MRI boundaries are noisy or partially missing. The method provides prior knowledge about the expected shape of an organ and reduces unrealistic segmentation results. Therefore, combining image information with PCA-based shape constraints can produce more reliable and anatomically meaningful segmentation.

Question 3: Machine Learning Methods – SVM, HMM, GMM and EM

Scenario

Recognize human activities such as walking, running, and jumping from video feature trajectories.

Answer

Understanding of Data:

The dataset consists of video sequences recorded at approximately 30–60 FPS. Each video may contain activities such as walking, running, and jumping. Feature trajectories can be represented using joint positions such as x and y coordinates or optical-flow vectors for each frame. Training labels identify the activity performed in each video. Synthetic CSV data containing simulated joint coordinates can also be used for experimentation.

Analysis:

Human activity recognition is a sequential classification problem because the movement changes over time. Different machine learning methods can be used for different aspects of the problem.

SVM (Support Vector Machine):

SVM can classify activities based on extracted features such as average joint movement, velocity, acceleration, or trajectory statistics. It works well when the feature vectors clearly separate different activities.

HMM (Hidden Markov Model):

HMM is suitable for modeling temporal sequences. The hidden states can represent stages of an activity, while observations represent joint positions or movement features.

GMM (Gaussian Mixture Model):

GMM models the probability distribution of feature vectors. It can represent multiple patterns within the same activity.

EM (Expectation-Maximization):

EM is an iterative optimization method commonly used to estimate the parameters of probabilistic models such as GMMs. It alternates between estimating the probability of data belonging to different components and updating the model parameters.

Application of Concepts:

The implementation can follow these steps:

1. Collect video sequences.
2. Extract frames from each video.
3. Extract joint coordinates or optical-flow features.
4. Normalize the extracted features.
5. Divide the data into training and testing sets.
6. Train an SVM classifier using extracted features.
7. Train an HMM to model temporal activity sequences.
8. Use GMM to model feature distributions.
9. Apply EM to estimate GMM parameters.
10. Predict the activity of unseen video sequences.
11. Compare predicted and actual activity labels.
12. Calculate accuracy, precision, recall, and F1-score.

Conclusion:

SVM is effective for feature-based activity classification, while HMM is particularly useful for modeling the temporal sequence of human movements. GMM and EM provide probabilistic modeling of movement patterns. Combining these approaches can improve activity recognition, particularly when activities have complex temporal variations.

Question 4: Deep Learning – CNNs, Transformers and GANs

Scenario

Detect rare obstacles in urban self-driving car images.

Answer

Understanding of Data:

The dataset contains RGB images of urban roads, typically around 1024×768 pixels. The images contain different objects such as vehicles, pedestrians, traffic signs, traffic cones, animals, and other obstacles. Bounding-box annotations are provided for the

target obstacles. The dataset may contain approximately 500–1000 labeled images. GAN-generated images can optionally be used to increase the amount of training data.

Analysis:

CNNs are highly effective for extracting spatial features from images. Early CNN layers identify low-level patterns such as edges and textures, while deeper layers identify complex structures and objects.

Transformers can model relationships between different regions of an image using attention mechanisms. This can help the model understand the relationship between an obstacle and its surrounding environment.

GANs can generate synthetic images that resemble real training samples. This is useful when rare obstacles have limited training examples. Additional synthetic samples can increase the diversity of the training data and reduce overfitting.

Application of Concepts:

The proposed workflow is:

1. Collect urban road images.
2. Resize and normalize the images.
3. Annotate obstacles using bounding boxes.
4. Divide the dataset into training, validation, and testing sets.
5. Apply image augmentation such as rotation, scaling, and cropping.
6. Train a CNN-based object detection model.
7. Use Transformer-based attention to capture long-range relationships.
8. Generate additional synthetic obstacle images using a GAN if required.
9. Combine real and suitable synthetic images for training.
10. Detect obstacles in unseen images.
11. Evaluate the model using precision, recall, F1-score and mean Average Precision (mAP).
12. Analyze false positives and false negatives.

Conclusion:

CNNs provide strong spatial feature extraction, Transformers help model relationships across image regions, and GANs can address the problem of limited rare-object training data. A hybrid deep-learning approach can therefore improve the detection of uncommon obstacles in autonomous driving environments. However, synthetic images should be carefully validated to ensure that they improve rather than reduce model performance.

Question 5: Graph-Based Learning for Shape Correspondence

Scenario

Find correspondences between 3D human face meshes.

Answer

Understanding of Data:

The dataset contains 3D human face meshes stored in OBJ or PLY format. Each mesh can contain approximately 5,000–50,000 vertices, and different meshes may have different numbers of vertices. The training dataset contains approximately 20–50 face meshes with annotated landmark correspondences. The test dataset contains approximately 5–10 unseen face meshes. Graph adjacency matrices may also be provided to describe connections between vertices.

Analysis:

A 3D face mesh can be represented as a graph in which each vertex represents a node and the connections between vertices represent edges. Graph-based learning methods can learn geometric relationships between different parts of the face.

Important facial landmarks such as the eyes, nose, mouth, and chin can be used as reference points. The model learns relationships between corresponding regions of different face meshes. Because different meshes may have different vertex counts, direct vertex-to-vertex matching may not be appropriate. Graph-based methods can instead use local neighborhood information and geometric features.

Application of Concepts:

The following steps can be used:

1. Load the 3D face meshes.
2. Extract vertices and edges from each mesh.
3. Construct graph representations.
4. Generate adjacency matrices.
5. Normalize the mesh coordinates.
6. Extract geometric features such as curvature and local neighborhoods.
7. Use annotated landmarks as training correspondences.
8. Train a graph-based correspondence model.
9. Compare features between vertices of different face meshes.
10. Establish corresponding points between the source and target meshes.
11. Test the model on unseen face meshes.

12. Evaluate correspondence accuracy using landmark distance or correspondence error.

For example, if the nose tip is identified as a landmark on one face, the model should identify the corresponding nose-tip location on another face despite differences in mesh resolution and facial geometry.

Conclusion:

Graph-based learning is well suited for 3D face correspondence because it preserves relationships between neighboring vertices and can handle meshes with different structures. Using landmarks and geometric features improves the reliability of correspondence. The effectiveness of the method can be evaluated using landmark correspondence accuracy and average geometric error.

Overall Comparative Conclusion

The five scenarios demonstrate different approaches for solving computer vision and pattern-recognition problems:

Question	Technique	Main Application	Suitable Evaluation
1	Hough Transform	Road, river and canal detection	Detection accuracy, false positives
2	PCA + Shape Priors	MRI organ segmentation	Dice, IoU, precision, recall
3	SVM, HMM, GMM, EM	Human activity recognition	Accuracy, precision, recall, F1-score
4	CNN, Transformer, GAN	Rare obstacle detection	mAP, precision, recall, F1-score
5	Graph-Based Learning	3D face correspondence	Landmark accuracy, correspondence error

Overall, the selection of the appropriate technique depends on the nature of the data and the problem. Hough Transform is effective for linear structures, PCA is useful for modeling shape variation, classical machine learning methods handle feature and temporal patterns, deep learning methods are powerful for complex image recognition, and graph-based learning is suitable for structured 3D data.